PhyDiff: Physics-Prior Diffusion Model for Cat's-Eye Pattern Identification From Satellite Imagery

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***Abstract*—Cat's-eye vortices appear in satellite observations as curved and spatially discontinuous filamentary patterns. Conventional region-based methods can identify the approximate vortex extent but fail to preserve filament connectivity and internal topology, while the scarcity of pixel-level annotations constrains data-driven learning. To address these challenges, we propose PhyDiff, a physics-prior diffusion model framework for cat's-eye vortex pattern identification. A cat's-eye simulation dataset (CATSIM) is first constructed from the analytical Stuart-vortex formulation, providing paired synthetic observations and topology-consistent masks for training. Within the reverse diffusion process, a Fourier residual block with frequency selection (FRB-FS) is introduced to enhance informative high-frequency components and preserve the continuity of thin vortex filaments. A contour-based geometric parameterization method is further developed to quantify vortex geometry without imposing circular or elliptical shape assumptions. Without additional fine-tuning, PhyDiff successfully extracts coherent cat's-eye filaments from Sentinel-2 observations over the Baltic Sea. The experimental results demonstrate that the proposed framework effectively preserves fine-scale filamentary structures and derives geometric parameters that are broadly consistent with the Stuart-vortex formulation. PhyDiff therefore provides a reproducible framework for the fine-scale morphological extraction and quantitative characterization of cat's-eye vortices from satellite remote sensing imagery.**

***Index Terms*—Cat's-eye vortex, diffusion model, image segmentation, mesoscale eddy, parameterization.**

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I. Introduction

OCEAN eddies are among the dominant coherent structures governing the transport, mixing, and redistribution of heat, momentum, and biogeochemical properties in the upper ocean [2]. Mesoscale eddies [3] generally have horizontal scales ranging from tens to hundreds of kilometers, although their characteristic dimensions vary substantially with latitude, stratification, and the local first baroclinic Rossby deformation radius [4]. Through water-mass exchange, lateral stirring, and vertical transport, ocean eddies can redistribute nutrients between the ocean interior and the euphotic zone, thereby influencing phytoplankton growth, marine ecosystem functioning [5], [6], [7], and upper ocean carbon cycling [8], [9]. Consequently, the reliable identification and characterization of ocean eddies have long remained central topics in physical oceanography and satellite remote sensing.

Among vortex-induced surface patterns, cat's-eye structures represent a distinctive filamentary morphology associated with nonlinear vortex organization in shear flows [10]. During their evolution, mesoscale eddies may develop regions of enhanced horizontal strain and velocity shear, which can promote secondary instabilities and reshape surrounding tracer distributions. These deformation processes can produce coherent recirculation regions and elongated filamentary structures embedded within or around the eddy [11], [12], [13], [14]. The classical Stuart solution provides a theoretical framework for describing this nonlinear vortex topology, characterized by closed recirculation regions, separatrix-like filaments, and hyperbolic stagnation points, as illustrated in Fig. 1. In the formulation, the dimensionless parameter α controls the amplitude and geometric development of the cat's-eye structure. However, the detailed filament geometry of satellite-observed cat's-eye patterns remains insufficiently quantified and the key parameter in the Stuart solution to be determined.

Traditional approaches for observing ocean eddies include in situ measurements [15], [16] and satellite-altimetry-based identification [17]. For example, Chen et al. [18] used orientation-aligned effective boundaries from nearly 2.6 million altimetry-derived eddies and showed that their climatological mean shape is oval and can be well approximated by an ellipse. These SSH-derived bulk geometries are compatible with complex tracer filaments. Under geostrophic balance, SSH contours approximate instantaneous surface streamlines,

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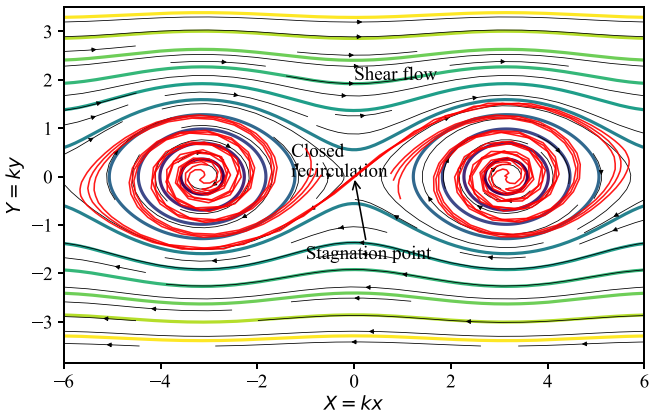


Fig. 1. Schematic of a cat's-eye vortex embedded in a background shear flow. The cat's-eye vortex is described by the Kelvin–Stuart solution with the steady stream function: $\psi(x, y) = -k^{-1}U \log[\cosh(ky) - \alpha \cos(kx)]$ [1], where k denotes the horizontal wavenumber associated with the periodicity of the vortex structure, U represents the background flow velocity, and α controls the degree of streamline distortion. The logarithmic term and prefactor $k^{-1}U$ are omitted since they do not affect the streamline topology. The same wavenumber k is used to normalize the spatial coordinates as $X = kx$ and $Y = ky$. Therefore, the axes in this figure represent nondimensional coordinates rather than physical distances. For $0 < \alpha < 1$, the flow exhibits alternating elliptic and hyperbolic critical points arranged periodically along the streamwise direction. Closed streamlines form around elliptic points, generating recirculation cells embedded within the background shear, while hyperbolic stagnation points organize material exchange through their stable and unstable manifolds.

whereas tracer patterns integrate finite-time advection, deformation, mixing, and source–sink processes. However, in situ observations are often spatially sparse [19], whereas gridded altimetry products have limited effective spatial resolution for resolving the narrow internal and peripheral filamentary structures associated with eddies. One way to overcome this obstacle is to utilize other satellite remote sensing data, such as synthetic aperture radar (SAR) and optical ocean-color observations, including chlorophyll [20], [21], which are available at high resolution and wide coverage. Nevertheless, a continuous tracer field does not inherently define an explicit cat's-eye object or a unique structural boundary. Delineation based on intensity or gradient thresholds depends on regional background conditions, image preprocessing, smoothing scales, and parameter selection, potentially fragmenting weak filaments or producing inconsistent results across scenes [22]. Therefore, accurate extraction and geometric characterization of cat's-eye pattern from remote sensing imagery is a prerequisite for further physical and dynamic analysis. The geographical distribution of the collected cat's-eye vortex observations in the Baltic Sea, together with their observation dates, is shown in Fig. 2.

Recent advances in deep learning (DL) have substantially improved automatic ocean eddy detection and segmentation from satellite imagery [23]. Zi et al. [24] and Li et al. [25] developed detection frameworks for locating eddies in SAR and multispectral imagery, whereas Saida et al. [26] and Hammoud et al. [27] further introduced semantic segmentation methods for delineating complete eddy regions, including eddies with irregular outer boundaries. These methods provide effective estimates of eddy locations and spatial extents. However, their outputs are primarily bounding boxes or region-level masks designed to represent the overall eddy region rather than its internal and peripheral filamentary organization. Although such masks can support estimates of the eddy center, they do not explicitly resolve the curved arms, internal openings, and separatrix-like filamentary boundaries that characterize cat's-eye morphology. These structures may be smoothed, merged, or left unresolved in a regional representation, leading to uncertainties in filament-specific quantities. Such quantities are required for reliable Stuart-model-based estimation of α, even when the overall eddy region is correctly identified. Consequently, existing eddy-detection and region-segmentation methods do not yet provide the filament-resolved representation required for quantitative characterization of observed cat's-eye patterns. The scarcity of reliable pixel-level annotations for weak, fragmented, and intertwined filaments further constrains the development of fully supervised models for this task.

To address these issues, we propose PhyDiff, a physics-prior diffusion model for cat's-eye vortex pattern identification from satellite imagery. Rather than inferring the underlying vortex dynamics from surface tracer patterns, our objective is to extract explicit morphological representations from remote sensing observations and provide quantitative descriptions of observed cat's-eye structures. Unlike CNN-based segmentation methods [28], [29], diffusion models have better generalization ability and improve the segmentation ability of realistic vortex images [30]. PhyDiff generates paired synthetic observations and topology-consistent masks from the analytical Stuart-vortex formulation, thereby introducing equation-guided morphological priors while reducing dependence on manual annotations. A Fourier-guided U-Net is embedded in the reverse diffusion process to enhance informative high-frequency features and preserve thin, weak, and continuous filaments. The predicted masks are subsequently processed using a contour-based parameterization method to estimate the geometric characteristics of the observed cat's-eye structures and derive the corresponding Stuart parameter. The main contributions are summarized as follows.

1) A physics-prior diffusion model segmentation framework is proposed for fine-scale extraction of cat's-eye vortex morphology from remote sensing imagery. Instead of relying solely on scarce manually labeled observations, the proposed framework generates synthetic image–mask pairs from a Stuart-vortex-based analytical formulation, thereby introducing equation-guided morphological priors for learning filamentary vortex-like patterns.
2) We introduce a Fourier residual block with frequency selection (FRB-FS) in the conditional diffusion model that amplifies informative high-frequency signals while suppressing noise, allowing for precise recovery of fine line structures and ensuring stable gradient propagation during the diffusion process, thus addressing the gap in vortex segmentation research.
3) A contour-based geometric parameterization strategy is introduced to quantitatively characterize segmented cat's-eye structures. By extracting morphology-related descriptors from the predicted contours, the proposed method establishes a quantitative link between

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satellite-observed filamentary morphology and the theoretical cat's-eye vortex model, providing a reproducible framework for morphological comparison and analysis.

II. Background and Data

A. Semantic Segmentation Models Based on Diffusion Model

Diffusion-based generative models [31], [32] have recently attracted increasing attention in semantic segmentation tasks due to their strong capability in modeling complex data distributions through iterative conditional denoising. Compared with conventional one-pass discriminative networks [33], diffusion models provide a natural framework for progressive structure reconstruction, which is particularly beneficial for capturing fine geometric details. Several studies have explored diffusion-based segmentation strategies. For example, Amit et al. [34] modeled segmentation masks as a denoising diffusion process conditioned on input imagery, enabling end-to-end training without relying on pretrained backbones, where image features and intermediate mask predictions are fused through dual-encoder representations. Similarly, Han et al. [35] proposed a two-stage unsupervised and supervised semantic segmentation framework using multiblur-based cold diffusion, which achieved an 85.8% mean intersection over union (mIoU) on the CRACK500 dataset. Zhang et al. [36] proposed a crack diffusion model that leverages the generative capabilities of diffusion models to capture the spatial distribution of cracks through iterative steps, achieving strong segmentation performance on public datasets. Other studies [37] further introduced cross-conditional encoder–decoder architectures and generative crack modeling strategies to improve spatial structure preservation. Although these approaches have improved segmentation performance, their applicability to highly sparse foreground structures, especially fine-grained curved line patterns, remains insufficiently explored. In this article, we introduce an FRB-FS module to enhance high-frequency structural information during the reverse diffusion process, thereby improving the reconstruction fidelity of sparse line structures.

B. Dataset Simulation

1) Governing Flow Field: As introduced in the opening section, the cat's-eye pattern streamline is constructed upon the classical Stuart solution of the 2-D incompressible equations, which admits steady nonlinear shear flows exhibiting characteristic cat's-eye streamline patterns. Its streamline function is expressed in simplified form as

$$\psi(x, y) = \cosh(ky) - \alpha \cos(kx) \tag{1}$$

and is used as a topology-generating representation of the Stuart contour family. Because the logarithmic transformation in the full Kelvin–Stuart streamfunction is monotonic, q preserves the same level-set topology. Here, k and α are treated as controllable structural parameters. In this study, we note that the velocity field is given by the spatial derivatives of the stream function

$$\mathbf{u}_{base} = (u, v) = \left(\frac{\partial \psi}{\partial y}, -\frac{\partial \psi}{\partial x}\right). \tag{2}$$

The resulting velocity field corresponds to the canonical Stuart cat's-eye structure prior to external modulation. No additional scaling is introduced at this stage; all velocity amplitudes are normalized for numerical integration.

2) Background Shear Embedding: Although the Stuart solution provides a morphologically consistent cat's-eye vortex, real oceanic vortices are rarely isolated. They are typically embedded within larger-scale background currents that introduce weak environmental shear. This shear primarily affects the outer streamlines, whereas the inner recirculation core remains relatively coherent.

To incorporate this environmental modulation without distorting the intrinsic cat's-eye pattern, a spatially weighted linear shear flow is superimposed onto the base field

$$\mathbf{u}_{shear}(x, y) = \beta y \left(1 - e^{-(r/r_s)^2}\right), \quad r = \sqrt{x^2 + y^2}. \tag{3}$$

where β denotes the background shear intensity and the term βy corresponds to a classical linear Couette-type shear flow. However, instead of applying it uniformly across the domain, a Gaussian-type radial modulation is introduced through the factor $1 - e^{-(r/r_s)^2}$. When near the vortex center ($r \to 0$), the imposed shear vanishes. Away from the core ($r \gg r_s$), the shear asymptotically approaches the linear profile.

Besides, r_s defines the radial transition scale separating the coherent vortex core from the environmentally influenced periphery.

3) Elliptic Envelope Constraint: Satellite SAR observations indicate that real cat's-eye vortices often exhibit elongated and elliptic outer envelopes rather than perfectly circular boundaries. The classical Stuart solution, however, is spatially periodic and does not naturally impose a finite geometric extent.

To reproduce the observed morphology while maintaining the internal streamline topology, a geometric soft confinement term is introduced

$$\mathbf{u}_{conf} = -k_c \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} - 1\right)_+ \left(\frac{2x}{a^2}, \frac{2y}{b^2}\right) \tag{4}$$

where a and b are the semi-major and semi-minor axes of the prescribed elliptic boundary, respectively; k_c controls the confinement strength; and $(\cdot)_+ = \max(\cdot, 0)$. The expression inside parentheses defines an elliptic level-set function. The confinement term is activated only when particles exceed the prescribed boundary ($x^2/a^2 + y^2/b^2 > 1$). The vector direction $(x^2/a^2, y^2/b^2)$ corresponds to the outward gradient of the elliptic function such that the restoring velocity always points inward. It is important to note that this term does not represent a physical body force. Instead, it acts as a geometric constraint.

The complete velocity field used for trajectory integration is therefore

$$\mathbf{u} = \mathbf{u}_{base} + \mathbf{u}_{shear} + \mathbf{u}_{conf} \tag{5}$$

which defines the overall vortex structure. Based on this field, a vortex mask $M(x, y)$ is obtained that represents the morphology of the vortices. This mask is then used to synthesize SAR images.

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TABLE I
KEY PARAMETERS USED IN THE SYNTHETIC CAT'S-EYE VORTEX DATASET GENERATION. THE TABLE LISTS THE PHYSICAL OR NUMERICAL ROLE OF EACH PARAMETER AND ITS TYPICAL RANGE

Parameter	Description	Range
k	vortex wavelength	0.8–1.0
β	background shear	0–0.008
a, b	elliptic axes	$a = 3.0$, $b = 1.5$
η	SAR contrast	0.3–0.6
α	Vortex amplitude	0.2–0.8
k_c	Elliptic confinement strength	0.7

4) SAR Forward Modeling: SAR is an active microwave remote sensing technique that transmits radar pulses toward the Earth's surface and records the returned backscattered signals. Over the ocean, the normalized radar cross section (NRCS) is primarily sensitive to small-scale sea-surface roughness, which can be modulated by wind, surface currents, convergence and divergence, and the redistribution of surface active materials. To establish a controlled link between vortex morphology and radar observations, a simplified SAR forward model is adopted. The NRCS in VV polarization is modeled as

$$\sigma^0_{VV}(x,y) = \bar{\sigma}_{VV}\left[1 + \eta\left(M(x,y) - 0.5\right)\right] \tag{6}$$

where $\bar{\sigma}_{VV}$ denotes the mean backscatter, $M(x,y) \in \{0,1\}$ represents the vortex mask, and η controls the vortex-induced contrast amplitude. This formulation assumes that the presence of the vortex modulates local surface roughness conditions, resulting in contrast variations in the observed backscatter. While simplified, this approach isolates geometric structure from complex air–sea interaction processes, enabling controlled evaluation of vortex detection and segmentation algorithms.

To account for the multiplicative nature of SAR speckle, the observed intensity is modeled as

$$I(x,y) = \sigma^0_{VV}(x,y)\, n(x,y) \tag{7}$$

where $n(x,y)$ is a Gamma-distributed random variable representing fully developed speckle.

To enhance structural diversity, key physical and imaging parameters—including k, β, elliptic axes (a,b), and contrast factor η—are randomly sampled within the ranges specified in Table I. Each simulation produces a paired SAR image and its corresponding vortex mask, forming the dataset used for training and evaluation.

III. METHODOLOGY

The objective of this article is to accurately extract the filamentary structures of the cat's-eye pattern and estimate their geometric characteristics from remote sensing imagery. Let $I \in \mathbb{R}^{H\times W\times 1}$ denote the input image and $M \in \{0,1\}^{H\times W}$ denote the corresponding vortex mask. The segmentation task can therefore be formulated as learning a mapping from the observed image I to the vortex structure M. To address the challenges posed by weak boundaries and complex filamentary patterns, we propose a physics-prior diffusion-based framework for vortex segmentation and parameterization. First, the segmentation problem is formulated as a conditional diffusion process, where a Fourier-guided U-Net architecture is employed to preserve high-frequency structures during iterative denoising. Finally, the segmented vortex contours are used to estimate geometric parameters that quantitatively characterize the cat's-eye morphology. The overall framework of the proposed segmentation method is illustrated in Fig. 3. In the following subsections, we describe each component in detail.

A. Diffusion Model of Segmentation

We design our model based on the diffusion model proposed in DDPM [38]. DDPMs define a forward diffusion process that gradually transforms data samples into Gaussian noise through a fixed Markov chain and learn a reverse denoising process that reconstructs the data distribution. In this work, the diffusion process operates on the segmentation mask. Specifically, Gaussian noise is progressively added to the clean mask x_0 over T diffusion steps, producing a sequence of latent variables $\{x_t\}_{t=1}^T$

$$q(x_t|x_{t-1}) = \mathcal{N}\left(x_t;\ \sqrt{1-\beta_t}x_{t-1}, \beta_t I\right) \tag{8}$$

where $\beta_t \in (0,1)$ follows a monotonically increasing variance schedule. The process admits a closed-form solution

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0,1) \tag{9}$$

where $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$. The reverse process aims to recover the clean segmentation mask by iteratively denoising x_t through a parameterized Markov chain

$$p_\theta(x_{t-1}|x_t, I). \tag{10}$$

By conditioning the reverse diffusion process on the input image, the model incorporates spatial constraints from the observation, enabling more accurate reconstruction of vortex structures.

B. Fourier Guided U-Shaped Network

The proposed Fourier-guided U-Net follows a symmetric encoder–decoder architecture tailored for conditional diffusion segmentation. At each diffusion timestep t, the noisy mask x_t is concatenated with the corresponding grayscale image I and passed through an initial convolution layer

$$F_0 = Conv([x_t \parallel I]). \tag{11}$$

The encoder consists of three hierarchical stages. Each stage contains two FRB-FS $\mathcal{R}_l(\cdot, e_t)$ followed by a downsampling operator $\mathcal{D}_l(\cdot)$

$$F_{l+1} = \mathcal{D}_l(\mathcal{R}_l(F_l, e_t)) \tag{12}$$

where e_t denotes the diffusion timestep embedding. The encoder progressively captures multiscale contextual features while maintaining awareness of the diffusion step.

At the bottleneck layer, the lowest resolution feature map F_{low} is further processed by residual blocks and a multihead self-attention module

$$B = A(\mathcal{R}_{bot}(F_{low}, e_t)) \tag{13}$$

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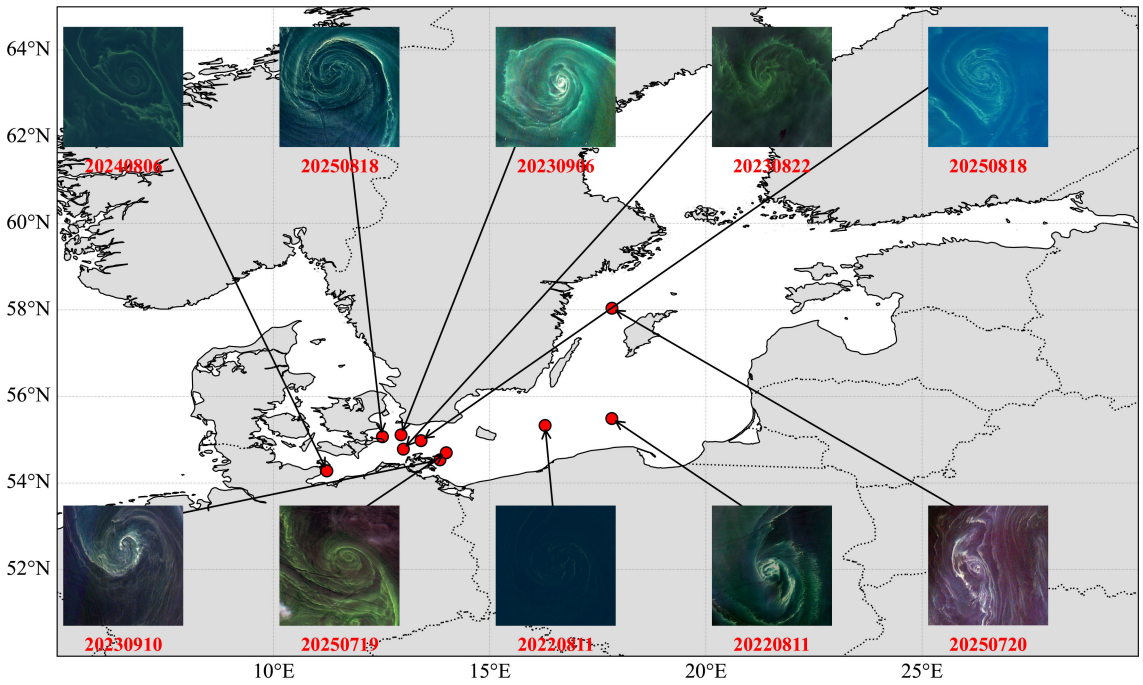


Fig. 2. Geographical distribution of the collected cat's-eye vortices in the Baltic Sea, together with their observation dates.

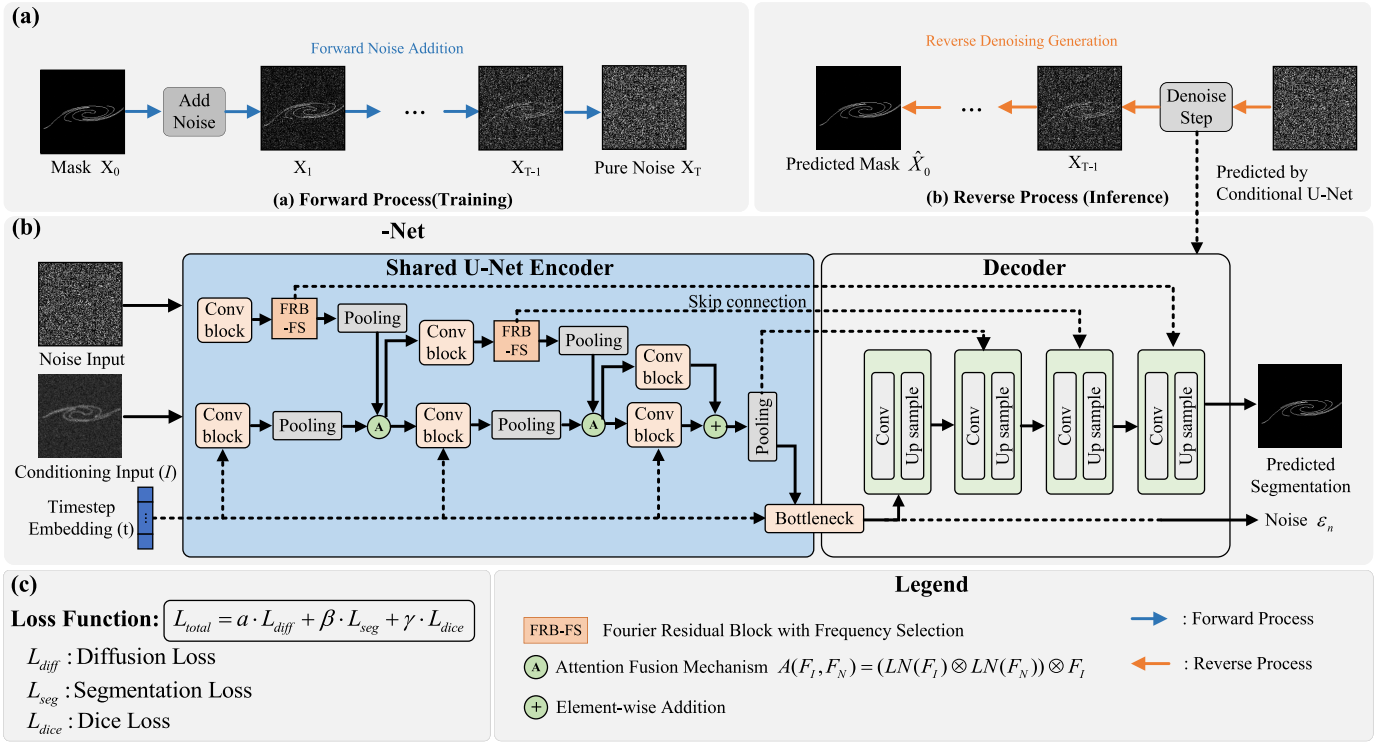


Fig. 3. Network architecture of the proposed diffusion-based segmentation model. (a) Forward and reverse diffusion process: (1) forward process (training) and (2) reverse process (inference). (b) Proposed conditional denoising U-Net. (c) Hybrid loss function and legend.

where $F_{low} = F_3$ is the feature map from the last encoder stage. The self-attention module captures long-range dependencies and global structural coherence.

The decoder progressively restores the spatial resolution. Starting from the bottleneck feature $G_3 = B$, each decoder stage performs upsampling, skip connection fusion with the corresponding encoder feature, and residual refinement

$$\mathbf{G}_l = \mathcal{R}'_l\left(\left[\mathcal{U}_l\left(\mathbf{G}_{l+1}\right) \;||F_l\right], e_t\right). \tag{14}$$

Finally, a 3 × 3 convolution generates the predicted clean mask

$$\hat{x}_0 = Conv_{out}\left(\mathbf{G}_0\right). \tag{15}$$

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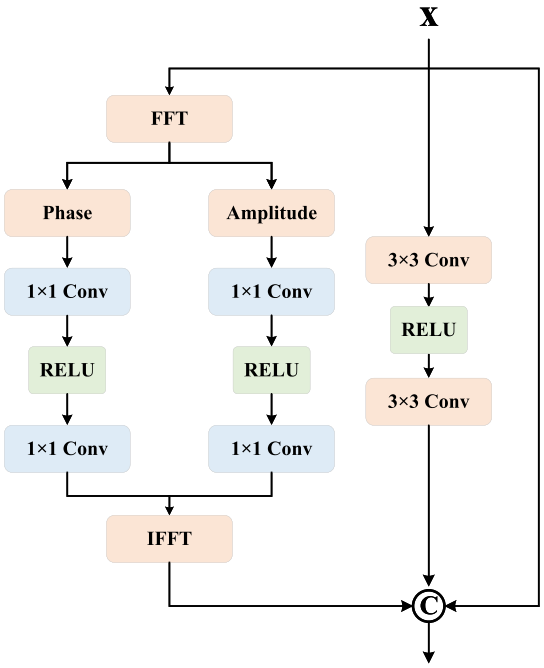


Fig. 4. Proposed FRB-FS.

C. FRB-FS

Diffusion-based segmentation models, while effective at modeling complex data distributions, the repeated downsampling operations in U-Net may suppress high-frequency components during feature extraction. In the task of segmenting the line structure, this limitation is particularly problematic.

To address this issue, we propose an FRB-FS, which explicitly enhances high-frequency representations while preserving spatial context. As shown in Fig. 4, given an input feature map $Z \in \mathbb{R}^{H\times W\times C}$ and a timestep embedding $e_t \in \mathbb{R}^{1\times d_t}$. Specifically, the input is normalized, activated, and passed through two 3×3 convolution layers, with the timestep embedding linearly projected and added to condition the block on the diffusion step

$$\hat{Z} = Conv_1\left(Act\left(Norm_1\left(Z\right)\right)\right) + MLP\left(e_t\right) \tag{16}$$

$$H_{spatial} = Conv_2\left(Act\left(Norm_2\left(\hat{Z}\right)\right)\right). \tag{17}$$

To recover fine line structures, $H_{spatial}$ is transformed into the frequency domain via a 2-D Fourier transform

$$F = FFT\left(H_{spatial}\right). \tag{18}$$

ReLU activations are applied separately to the real and imaginary components to emphasize informative high-frequency signals while suppressing low-frequency noise

$$\begin{aligned} Z_{real} &= ReLU\left(Re\left(F\right)\right) \\ Z_{img} &= ReLU\left(Im\left(F\right)\right). \end{aligned} \tag{19}$$

These components are processed by 1×1 convolutions and reconstructed via inverse Fourier transform

$$H_{freq} = IFFT\left(Conv_{real}\left(Z_{real}\right) + i\cdot Conv_{img}\left(Z_{img}\right)\right). \tag{20}$$

Finally, a 3×3 convolution refines the spatial representation of the frequency branch, and a residual shortcut from the input is added to form the block output

$$FRB-FS\left(Z, e_t\right) = H_{spatial} + H_{freq} + ResConv\left(Z\right). \tag{21}$$

By integrating spatial and frequency branches with a residual connection, FRB-FS selectively amplifies high-frequency features while maintaining global context and stable gradient propagation.

D. Cross-Modal Generalization and Geometric Characterization

Unlike the approach in [24], which approximates eddies as simple ellipses, this study avoids such geometric assumptions. The cat's-eye pattern investigated exhibits complex filamentary structures and eye-shaped patterns that cannot be adequately represented by a simple ellipse. We develop a contour-based geometric parameterization framework that directly extracts meaningful eddy parameters from the segmented vortex mask.

After diffusion-based segmentation, a postprocessing stage is applied to derive geometric parameters from the predicted binary mask, without introducing additional learnable components. Specifically, the segmented vortex region $\Omega \subset \mathbb{R}^2$ is obtained by thresholding the predicted probability map. To ensure topological consistency, morphological closing is applied to close small gaps and remove spurious holes, and the external contour with the maximum arc length is selected as the vortex boundary. This contour is represented as an ordered set of points

$$Y = \left\{p_i \in \mathbb{R}^2 \middle| i = 1, \ldots, N\right\}. \tag{22}$$

This contour serves as the sole input for subsequent geometric analysis.

The geometric center of the vortex is defined as the centroid of the contour points

$$c = \frac{1}{N}\sum_{i=1}^{N} p_i. \tag{23}$$

To estimate the dominant orientation of the vortex, principal component analysis (PCA) [39] is performed on the centered contour points $p_i - c$. The covariance matrix is

$$\Sigma = \frac{1}{N}\sum_{i=1}^{N}\left(pi - c\right)\left(pi - c\right)^T. \tag{24}$$

The eigenvector corresponding to the largest eigenvalue defines the major-axis direction v_{maj}, capturing the vortex's length.

Rather than relying on bounding boxes or ellipse fitting, the major-axis length is measured as a chord strictly confined within the segmented region. A line passing through the centroid and oriented along v_{maj} is defined as

$$\ell_{maj}\left(t\right) = c + tv_{maj}, \quad t \in \mathbb{R}. \tag{25}$$

The intersections between this line and the contour are computed, and the two extreme points along v_{maj} are selected,

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giving a chord fully enclosed by the vortex region. The major-axis length is then defined as

$$L_{maj} = \left\| p^{+}_{maj} - p^{-}_{maj} \right\|_2. \tag{26}$$

The width direction is defined as the unit vector orthogonal to the major axis

$$v_{wid} = \begin{bmatrix} -u_{maj}, y \\ u_{maj}, x \end{bmatrix}, \quad \|v_{wid}\|_2 = 1. \tag{27}$$

A simple width estimate is obtained by projecting the contour points onto v_{wid}

$$W_{proj} = \max_i (p_i \cdot v_{wid}) - \min_i (p_i \cdot v_{wid}). \tag{28}$$

While this is easy to compute, it may overestimate the true internal width for concave or irregular vortex boundaries.

To obtain a reproducible geometric descriptor width strictly confined within the vortex region, a contour-clipped chord is employed. A line passing through the centroid and oriented along v_{wid} is defined as

$$\ell_{wid}(t) = c + t \cdot v_{wid}. \tag{29}$$

The intersections between ℓ_{wid} and the contour are computed, and the two extreme intersection points along v_{wid} are selected. The resulting chord lies entirely within Ω, and the true vortex width is defined as

$$W_{true} = \left\| p^{+}_{wid} - p^{-}_{wid} \right\|_2. \tag{30}$$

By construction, W_{true} avoids overestimation caused by concavities or nonelliptical geometries and provides a robust measure of the vortex's transverse scale.

This postprocessing strategy relies solely on the geometric properties of the segmented contour and introduces no assumptions about vortex shape. Unlike ellipse fitting or bounding-box-based approaches, the contour-clipped chord formulation guarantees that both the length and width measurements are physically consistent and fully contained within the detected vortex region.

IV. Experimental Results

A. Experimental Datasets' Description

To evaluate the proposed PhyDiff, we use a self-constructed cat's-eye simulation dataset (CATSIM), which is employed for model training and evaluation and serves as data support for subsequent vortex segmentation experiments on real remote sensing imagery. The CATSIM dataset comprises 3000 images of 256 $\times$256 pixels, each containing the VV polarization channel. Additionally, we collected a set of 30 real Cat's-eye structure images from Sentinel-2 Level products covering the period 2022–2025. These images were acquired over the Baltic Sea region and include multispectral bands at 10-m spatial resolution. All images retain precise georeferencing information, enabling accurate localization of vortex structures. This dataset serves to evaluate the generalization capability of the proposed PhyDiff framework under realistic observational conditions. All data are publicly available and can be downloaded from https://browser.dataspace.copernicus.eu/

B. Experimental Settings

In our experiments, all images were uniformly resized to 256 $\times$ 256 pixels. The model was trained for 100 epochs using the AdamW optimizer [40] with a learning rate of 1×10^{-4} and a batch size of 16. Inference was performed with ten diffusion steps, which is considerably fewer than in most previous studies [40], [41]. Experiments were implemented on the PyTorch platform and executed on an NVIDIA 5080 GPU. The dataset was split such that 70% of the images were used for training, 10% for validation, and the remaining 20% for testing. To evaluate the effectiveness of the proposed framework and other existing model performance, four commonly used quantitative analysis metrics are employed: precision, recall, $F1$-score, and mIoU.

C. Comparative Experiments

To evaluate the effectiveness of the proposed method, we compared it with several image segmentation models, including MiT-UNet [42], CT-crackseg [43], and CrackFormer [44], whose morphological characteristics are similar to the filamentary patterns studied in this work. MiT-UNet uses a U-Net-like encoder–decoder architecture and an efficient self-attention module to capture global contextual information and enhance the extraction of thin linear structures. CT-Crackseg employs dilated residual blocks to capture multiscale crack details and a boundary awareness module to emphasize crack boundary features. The network model in [44] adopts transformer encoder blocks with relative positional embeddings to capture long-range contextual dependencies and improve crack representation. In addition, a diffusion-based segmentation model [34] and a classical convolutional segmentation baseline [45] are also included for comparison.

As shown in Table II, the proposed PhyDiff method delivers outstanding segmentation performance in terms of precision, recall, $F1$-score, and mIoU across datasets, surpassing the other methods. This superior performance can be attributed to PhyDiff's ability to effectively capture interactions between cross-modal features and leverage prior knowledge. Furthermore, the proposed model maintains a moderate computational complexity, which is lower than Segdiff and CrackFormer while providing superior segmentation performance.

For a more intuitive visual evaluation, we visualized the segmentation results of each method. As shown in Fig. 5, conventional segmentation models such as U-Net tend to produce fragmented or oversmoothed predictions, failing to preserve the continuity and closure of fine-scale line structures. Although MiT-UNet is effective for thin line structures such as power lines, its design primarily enforces long-range consistency along a dominant direction, which is insufficient for capturing the highly curved and closed topology of cat's-eye structures. CT-CrackSeg is tailored for crack-like objects with strong local contrast and explicit boundaries, whereas cat's-eye patterns correspond to smooth regions without well-defined physical boundaries. Moreover, CT-CrackSeg favors locally clustered, block-wise responses rather than globally continuous curves. Notably, the proposed model preserves structural connectivity and suppresses

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TABLE II
COMPARATIVE RESULTS OF VARIOUS MODELS ON THE CATSIM DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Method	Params (M)	FLOPs (G)	CATSIM dataset			
			Precision (%)	Recall (%)	F1-score (%)	mIoU (%)
Unet	31.04	218.95	71.43	69.4	70.39	53.53
MiT-Unet(b1)	19.27	73.25	61.69	55.36	58.96	43.96
Segdiff	10.57	41.83	59.84	53.48	62.39	44.12
CrackFormer	**7.35**	**67.2**	71.68	71.49	71.56	55.45
CT-crackseg	23.98	81.03	68.19	64.21	78.23	64.33
Our	13.25	76.08	**79.79**	**79.93**	**80.57**	**66.97**

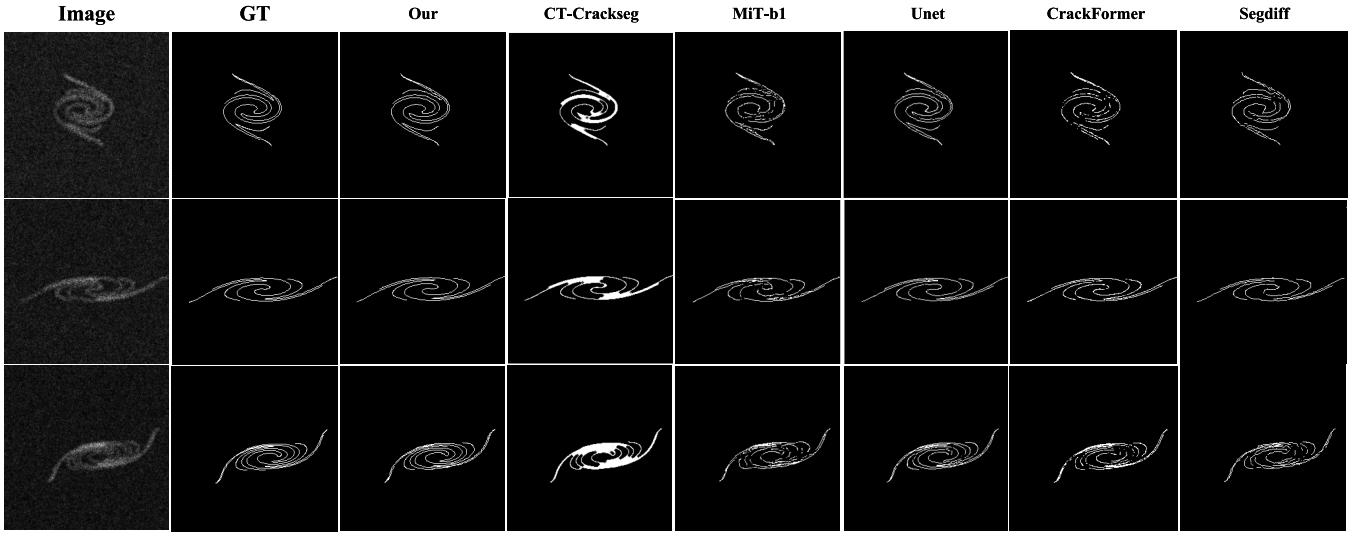


Fig. 5. Comparison of visualized experimental results on the CATSIM datasets.

TABLE III
ABLATION EXPERIMENTAL RESULTS FOR SELF-ATTENTION AND RES-FFT ON THE CATSIM DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Case	+Self-Attention	+FRB-FS	Precision(%)	Recall(%)	F1-score (%)	mIoU (%)
1	×	×	65.81	74.17	69.68	53.97
2	✓	×	66.11	74.65	70.06	63.12
3	×	✓	76.55	78.75	77.63	63.43
4	✓	✓	**79.79**	**79.93**	**80.57**	**66.97**

background noise more effectively, even though it has never been exposed to real images during training.

D. Ablation Experiments

We conducted ablation experiments to investigate the impact of various components, as detailed in Table III. Introducing the self-attention module at the bottleneck layer of the diffusion U-Net backbone, the overall performance improves slightly, with the $F1$-score increasing from 69.68% to 70.06%. More importantly, mIoU increases from 53.97% to 63.12%. This mechanism leverages the compressed latent representation to integrate spatially distant features, allowing the model to capture long-range dependencies and global contextual information, enhancing the structural consistency of the reconstructed mask during the decoding process. In Case 3, the FRB-FS module is incorporated, the precision increases to 76.55%, and the mIoU reaches 63.43%. This improvement can be attributed to the explicit enhancement of high-frequency components associated with line structures. When both components were employed simultaneously, the model achieved the highest performance, indicating that both components contribute positively to the overall segmentation quality.

In addition, we also evaluate the effectiveness of the proposed CATSIM dataset through a transfer experiment using the CRACK500 dataset [46], a publicly available benchmark for pavement crack segmentation. CRACK500 dataset contains 500 high-resolution road images with manually annotated crack masks and is widely used for evaluating the segmentation of thin, sparse, and elongated foreground structures. We selected this dataset because crack segmentation shares several visual characteristics with the proposed task, namely the extraction of narrow filamentary structures embedded in

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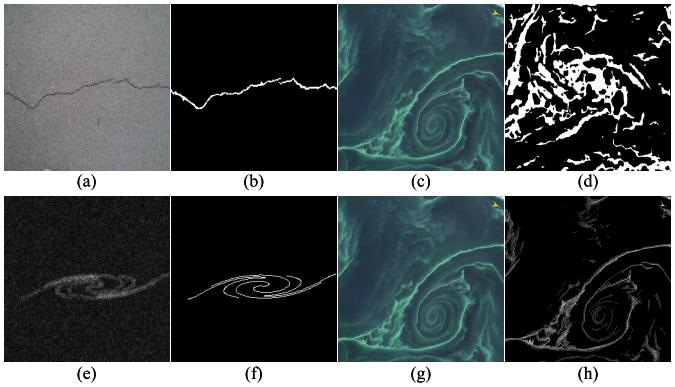


Fig. 6. Comparison illustrating the effectiveness of physics-based simulated training data. (a) and (b) Crack image and corresponding annotated crack label from the CRACK500 dataset used for training. (c) Sentinel-2 cat's-eye vortex image. (d) Segmentation result obtained using the model trained on the CRACK500 dataset. (e) and (f) Simulated Stuart vortex used as the proposed physics-based training sample and label. (g) Same image as in (c). (h) Segmentation result obtained using the model trained on the proposed simulated dataset.

complex backgrounds under severe foreground–background imbalance. As illustrated in Fig. 6, although the CRACK500 dataset contains sparse line structures, their geometric organization differs substantially from the filamentary patterns of cat's-eye eddies.

Consequently, the model trained on the CRACK500 dataset primarily learns generic local line features but lacks the ability to recover the coherent spiral topology characteristic of cat's-eye vortices. As shown in Fig. 6(d), the predicted filamentary structures are fragmented and fail to preserve the overall vortex organization. In contrast, the model trained on the proposed CATSIM dataset produces more complete and coherent vortex filaments, as shown in Fig. 6(h). This improvement indicates that the simulated dataset not only provides thin filamentary structures but also preserves the characteristic topological organization of cat's-eye vortices, enabling the network to learn physically meaningful structural priors rather than generic line patterns.

E. Qualitative Evaluation on Real Satellite Imagery

The practical applicability of the proposed method is assessed by directly applying the model, trained solely on the simulated CATSIM dataset, to real satellite observations. In realistic ocean environments, cat's-eye patterns often consist of numerous thin filaments and irregular structures, which makes accurate manual annotation challenging. Fig. 7 presents representative segmentation results on real observations. The first and third rows show the original observations, while the second and fourth rows display the corresponding predictions generated by the proposed model. Despite being trained only on simulated data, the model visually captures the filamentary structures of the cat's-eye pattern and preserves reasonable spatial continuity. These results indirectly demonstrate the effectiveness of the simulated dataset, which provides useful structural priors for segmenting cat's-eye vortices and facilitates the study of fine-scale vortex structures in real oceanic environments.

F. Cross Verification With Multisource Data

To verify that the detected "cat's-eye" patterns correspond to real physical vortices rather than algorithmic artifacts or noise, chlorophyll-a concentration data are used as an independent validation source, as shown in Fig. 8. The chlorophyll product is geographically coregistered with the satellite imagery in longitude and latitude, enabling direct spatial correspondence between the two data. Fig. 8 presents the enlarged regions of interest (ROIs) from both the satellite imagery and the corresponding chlorophyll fields for detailed comparison. As shown in the ROIs, coherent spiral and filamentary signatures are consistently observed in the chlorophyll fields, matching the fine-scale vortex structures detected in the satellite imagery. These consistent morphological characteristics provide independent evidence that the identified cat's-eye patterns correspond to real physical vortices rather than algorithmic artifacts.

G. Results in Vortex Parameters

Based on the segmentation results, the observed cat's-eye vortices are further analyzed to derive their structural parameters and morphology-based geometric descriptors. Representative results are shown in Fig. 9. The segmented vortex structures are first paired with their corresponding original satellite images. Following the procedure described in Section III, the detected vortex boundaries are delineated on the original images using green contours. The blue lines indicate the predicted vortex major-axis lengths and correspond to the major-axis directions of the vortices. The purple lines represent the predicted vortex minor-axis widths, oriented perpendicular to the length axes, and characterize the transverse scales of the vortices. In addition, the yellow points mark the predicted vortex centers, indicating the locations of the vortex cores. The extracted vortex parameters reveal clear geometric characteristics of the observed cat's-eye structures. As shown in Fig. 10, the detected vortices exhibit lengths ranging from approximately 7.1 to 20.4 km, while their widths vary between 3.2 and 11.9 km.

1) Consistency With the Stuart Cat's-Eye Vortex: To quantify the horizontal geometry of the extracted cat's-eye structures, the minor-to-major axis ratio (MAR) is defined as

$$\mathrm{MAR} = \frac{W}{L} \tag{31}$$

where L denotes the longitudinal length measured along the principal orientation of the cat's-eye structure and W denotes the transverse width measured perpendicular to this orientation. Both quantities are obtained from the contour-based geometric parameterization rather than from elliptical fitting. Under this definition, a smaller MAR indicates a more elongated cat's-eye structure.

Previous studies typically estimated the Stuart parameter α by fitting theoretical streamfunction contours to observed vortex boundaries, which may introduce additional empirical uncertainty [47]. In contrast, we relate the observed geometric parameters to the classical Stuart cat's-eye solution. In this solution, the streamfunction is given by

$$q = \cosh(ky) - \alpha\cos(kx), \quad k = 2\pi/L. \tag{32}$$

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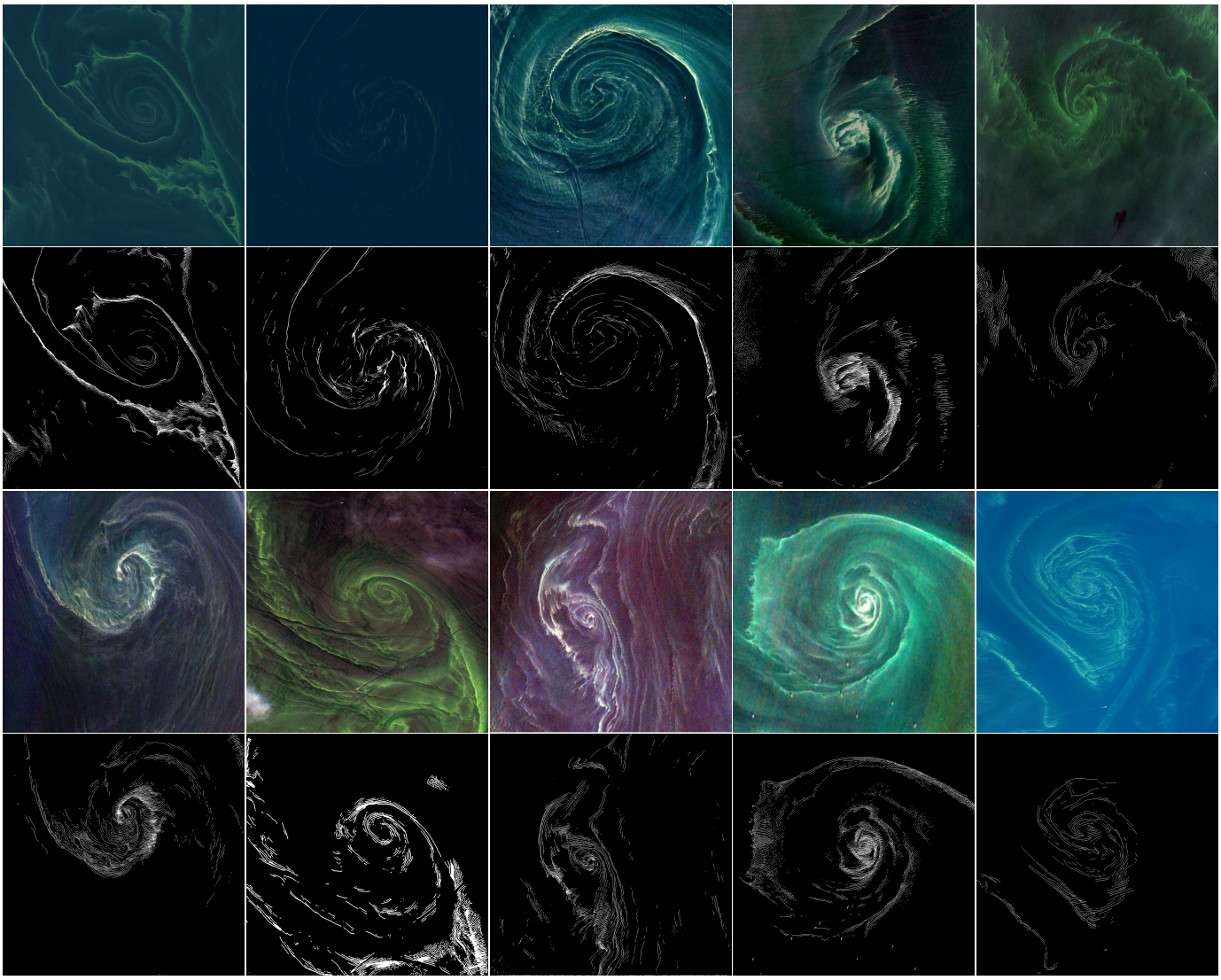


Fig. 7. Part of visualization results of segmentation for cat's-eye observations vortices.

Applying the streamline closure condition at $(x = 0, y = W/2)$ yields

$$\alpha = \frac{\cosh(\pi W/L) - 1}{2} \tag{33}$$

and this expression establishes a direct mapping between the observable geometric structure of the vortex and the theoretical parameter governing the cat's-eye flow topology. Furthermore, the Stuart solution requires $\alpha < 1$, and the vortex geometry must satisfy

$$MAR = W/L < 0.56. \tag{34}$$

This result provides a theoretically derived criterion for evaluating the geometric consistency of an extracted cat's-eye structure with the admissible parameter range of the idealized Stuart solution. The previously reported empirical value of 0.5 ± 0.2 [47] provides an observational reference for the width-to-length ratio, whereas the present result provides an analytical upper bound derived from the Stuart formulation.

Based on the derived Stewart solution constraints, the MAR should be $W/L < 0.56$. As shown in Fig. 10, the majority of the samples comply with this theoretical bound. However, two cases (Eddies 2 and 7) slightly exceed the threshold. These deviations may originate from uncertainties in the geometric extraction process. Overall, the statistical results demonstrate that the geometric properties of the detected vortices are generally consistent with the theoretical constraint, while the few outliers highlight potential limitations of the current parameterization and indicate directions for further model refinement.

2) Vortex Center Locations and Spatial Distribution: To quantitatively evaluate the localization accuracy of the proposed vortex center extraction method, we compared the estimated vortex centers with manually recorded reference positions. It should be noted that there is currently no publicly available dataset providing ground-truth vortex center coordinates for the studied events. Therefore, the reference coordinates were manually recorded during the visual inspection process when retrieving the corresponding satellite imagery from the Copernicus data portal. The localization

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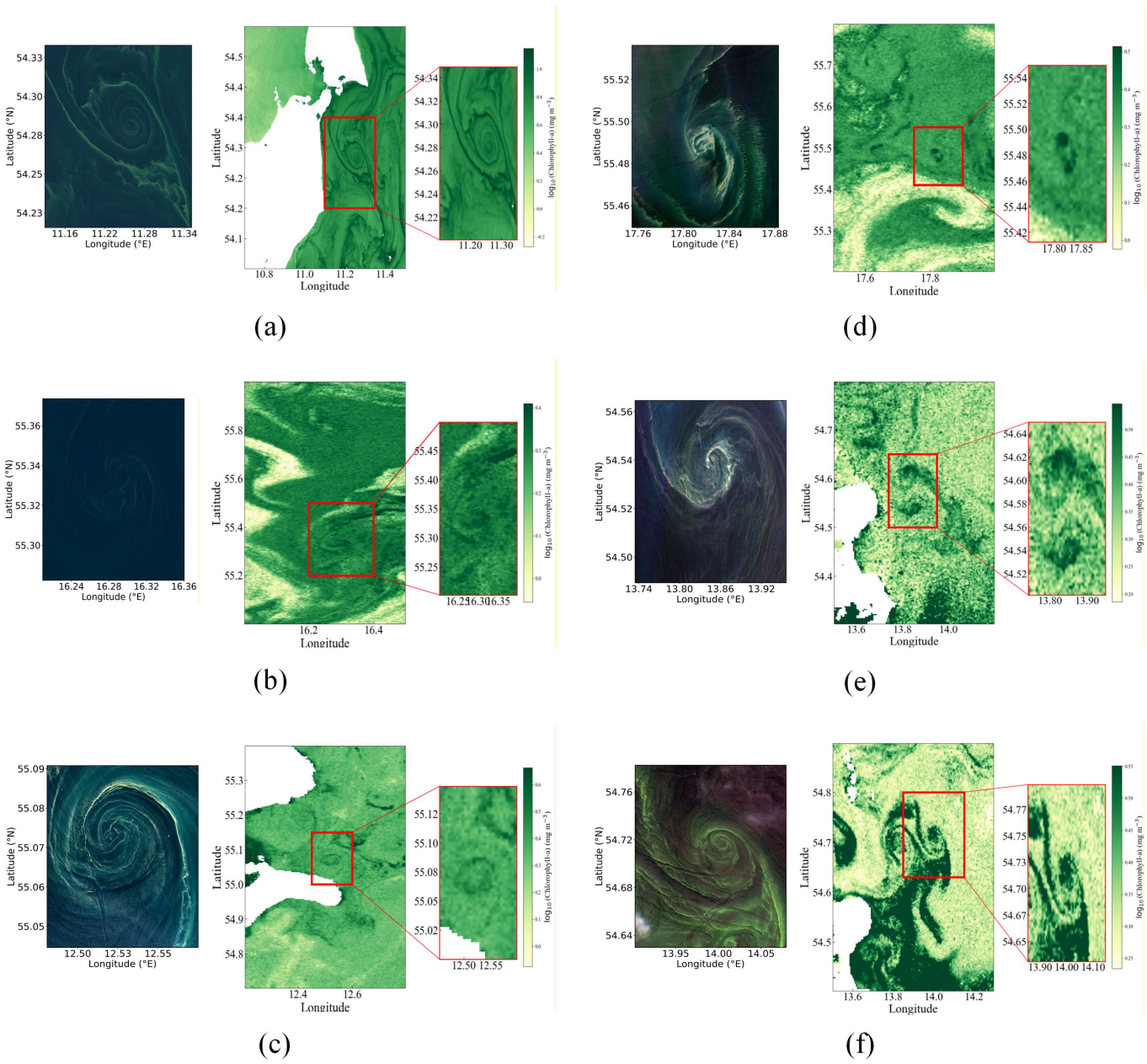


Fig. 8. Geolocation-consistent qualitative comparison between Sentinel-2 cat's-eye filamentary patterns and collocated chlorophyll-a observations. (a) August 6, 2024. (b) August 11, 2022. (c) August 18, 2025. (d) August 11, 2022. (e) September 10, 2023. (f) July 19, 2025.

error between the predicted and reference vortex cores was computed using geodesic distances on the WGS84 ellipsoid.

Fig. 11(a) illustrates the error vector distribution for ten vortex samples. The ground-truth vortex centers are aligned at the origin, while the predicted centers are represented as offset points. The arrows denote the error vectors from the ground truth to the predicted locations. The results show that most predicted vortex centers cluster close to the origin, indicating small spatial deviations. Reference circles corresponding to 500 m, 1 km, and 2 km are provided to illustrate the error magnitude.

Quantitatively, the proposed parameterization method achieves a mean deviation from the manually interpreted apparent centers of 1058.44 m, with a maximum error of 2786.03 m. Considering that the typical spatial scale of the identified cat's-eye vortices ranges from approximately 7–20 km, the localization error remains relatively small compared with the overall vortex size. In most cases, the positional deviation is within a small fraction of the vortex diameter. These results indicate that the proposed method can reliably estimate vortex core locations and provide sufficiently accurate geometric parameters for subsequent vortex characterization and analysis.

3) Idealized Stuart-Model Morphology Generated From the Extracted Geometric Descriptors: To examine how the extracted geometric descriptors map onto the adopted Stuart morphology, idealized contour patterns were generated using

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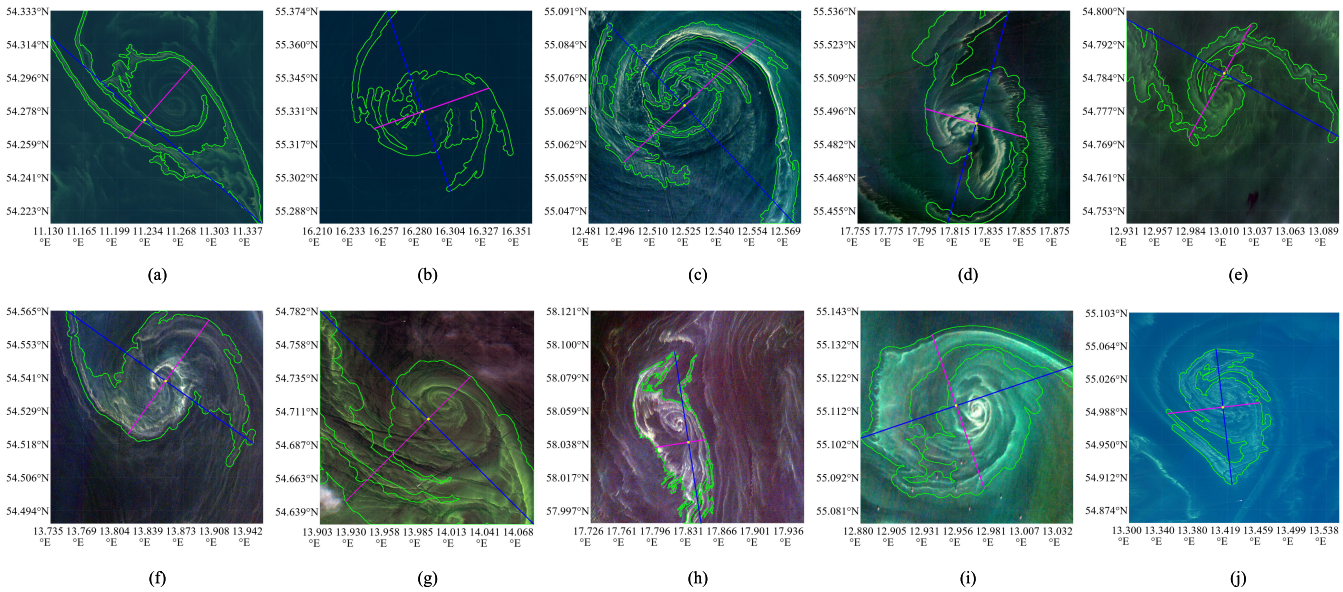


Fig. 9. Representative parameterization results for cat's-eye vortices. Panels (a)–(j) correspond, respectively, to the ten cat's-eye vortex cases shown in Fig. 7. The green contours indicate the detected vortex boundaries, blue lines represent the predicted major axes, purple lines represent the predicted minor axes, and yellow points indicate the predicted vortex centers.

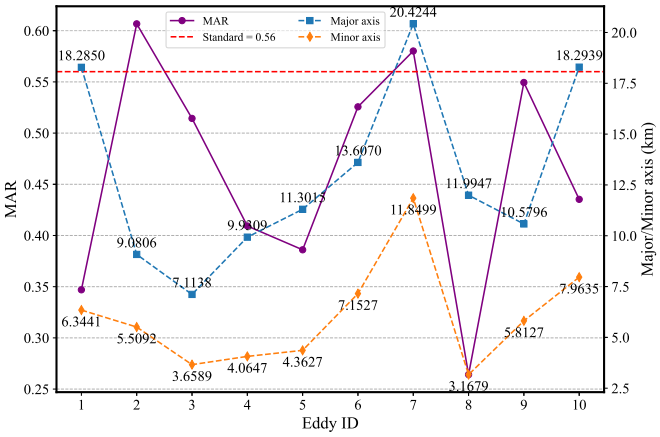


Fig. 10. MAR, minor-axis, and major-axis length of ten parameterized vortices. The red dashed line denotes the reference MAR 0.56 for cat's-eye vortices.

the estimated morphology-based Stuart equivalent parameter α and the corresponding geometric scales. The resulting contours are shown in Fig. 12. Smaller values of α generate elongated and flattened vortex structures, whereas larger values produce more symmetric and well-developed recirculation cells. Within the idealized Stuart model, these differences correspond to variations in the relative prominence of the recirculation cells and background shear.

For cases (b) and (g), the estimated values of α exceed the admissible range of the Stuart formulation adopted in this study. These cases cannot be represented as regular closed cat's-eye solutions within the current prior. For the remaining cases, the extracted geometric descriptors fall within the admissible domain of the Stuart-type mapping. This result indicates internal geometric compatibility between the extracted morphology and the adopted analytical prior and supports the model consistency of the geometric parameterization of the proposed parameterization framework.

4) Multisource Association and Case-Based Geometric Characterization: To demonstrate the applicability of the proposed framework to real remote sensing observations, this article selected a representative case that links the AVISO mesoscale eddy trajectory product, multidate Sentinel-2 observations, and the subsequent PhyDiff-based morphological analysis. Fig. 13(a) shows the selected AVISO-derived eddy trajectory from July 16 to 26, 2025. The upper subpanel provides the regional geographic context, whereas the lower subpanel enlarges the selected trajectory and displays the daily eddy-center positions. These center locations are directly provided by the AVISO mesoscale eddy trajectory product and are derived from the sea-surface height field; they are not estimated by the proposed PhyDiff framework. Fig. 13(a) therefore provides an SSH-based reference for the mesoscale location and displacement of the tracked eddy.

Fig. 13(b) presents the available Sentinel-2 observations acquired on July 15, 17, 24, 25, and 27, 2025. Because cloud-free optical observations are not available at daily intervals. For each optical image, the apparent center and radius were manually interpreted from the visible spiral and filamentary organization. The apparent center denotes the visually identified center of the recirculation-like pattern, whereas the apparent radius describes the approximate spatial extent from this center to the outer coherent filament. The multidate images show substantial changes in the visible filamentary morphology. The interpreted apparent radius increases from 11.85 km on July 15 to 23.25 km on July 25 and subsequently decreases to 11.16 km on July 27. However, these variations describe changes in the satellite-observed surface expression rather than the complete dynamical evolution of the eddy. The apparent morphology may also be influenced by

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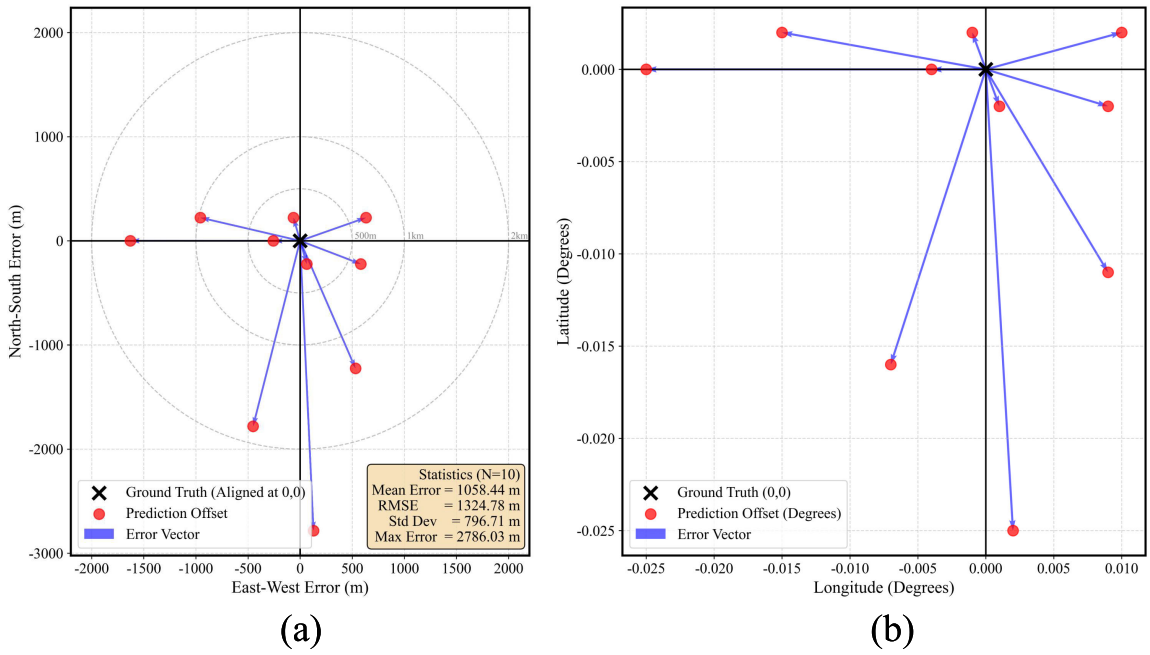


Fig. 11. Localization differences between the PhyDiff-derived morphological centers and the manually interpreted apparent morphological centers in Sentinel-2 imagery. (a) Planar center displacements after translating each manually interpreted reference center to the origin. The red circles denote the PhyDiff-derived morphological centers, the black cross at (0, 0) denotes the corresponding manually interpreted apparent morphological center, and the blue line segments represent the displacement vectors between the two center estimates. Summary statistics of the displacement magnitude, including the mean error, RMSE, standard deviation, and maximum error are also reported. (b) Signed geographic offsets of the PhyDiff-derived morphological centers relative to the manually interpreted reference centers, expressed as longitudinal and latitudinal differences in degrees. The manually interpreted centers are used only as image-based morphological references and should not be regarded as independently validated dynamical eddy centers.

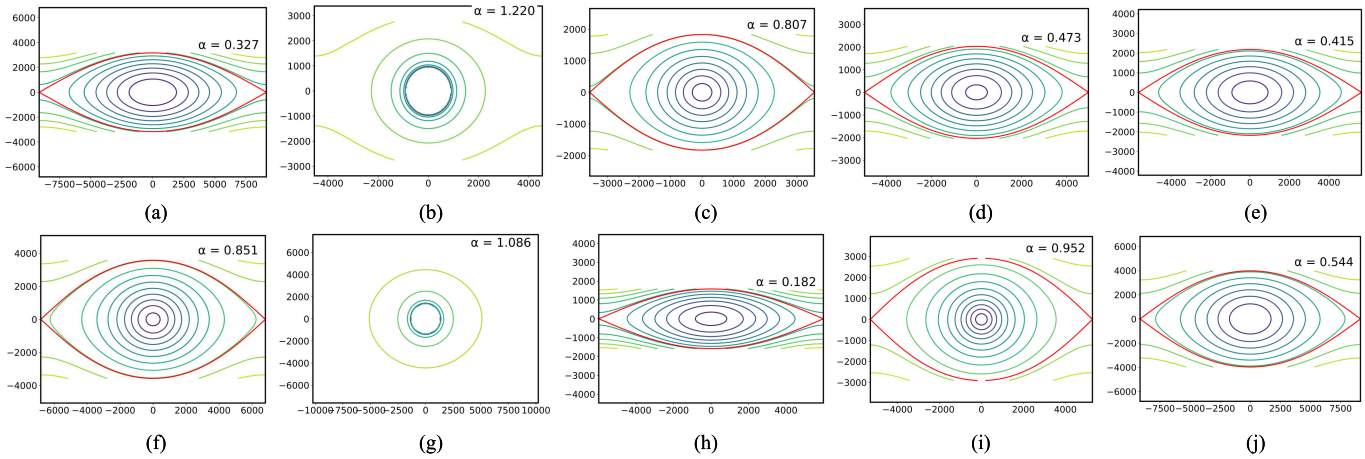


Fig. 12. Idealized Stuart contour patterns generated from the extracted geometric parameter α. The red curves represent specific boundaries or critical level sets within each contour map. (a)–(j) Contour lines for different values of the parameter α. The specific values are (a) $\alpha = 0.327$, (b) $\alpha = 1.223$, (c) $\alpha = 0.807$, (d) $\alpha = 0.473$, (e) $\alpha = 0.415$, (f) $\alpha = 0.865$, (g) $\alpha = 1.086$, (h) $\alpha = 0.182$, (i) $\alpha = 0.952$, and (j) $\alpha = 0.544$. (b) and (g) $\alpha > 1$, fall outside the admissible parameter range of the adopted Stuart prior and are shown only as prior-inconsistent mappings. The contours provide a visualization of geometric compatibility with the analytical prior.

tracer distribution, biological variability, cloud contamination, illumination conditions, and differences in image availability.

Moreover, Fig. 13(c) illustrates the PhyDiff-based analysis of the July 15 observation. The input image is the same cropped region shown in Fig. 13(b), thereby maintaining direct spatial correspondence between the optical observation and the model analysis. PhyDiff first converts the input image into a binary mask representing the dominant filamentary structure. The resulting mask preserves the principal spiral and curved filaments while suppressing most unrelated background responses. The predicted mask is then processed using the contour-based geometric parameterization procedure described in Section III-D and Algorithm 1. The dominant predicted contour is used to estimate the morphological center, principal orientation, longitudinal extent, and transverse width. For this case, the model-estimated center is located at approximately 26.02°E and 59.81°N. The estimated longitudinal extent is 24.64 km, and the transverse width is 9.23 km. The manually

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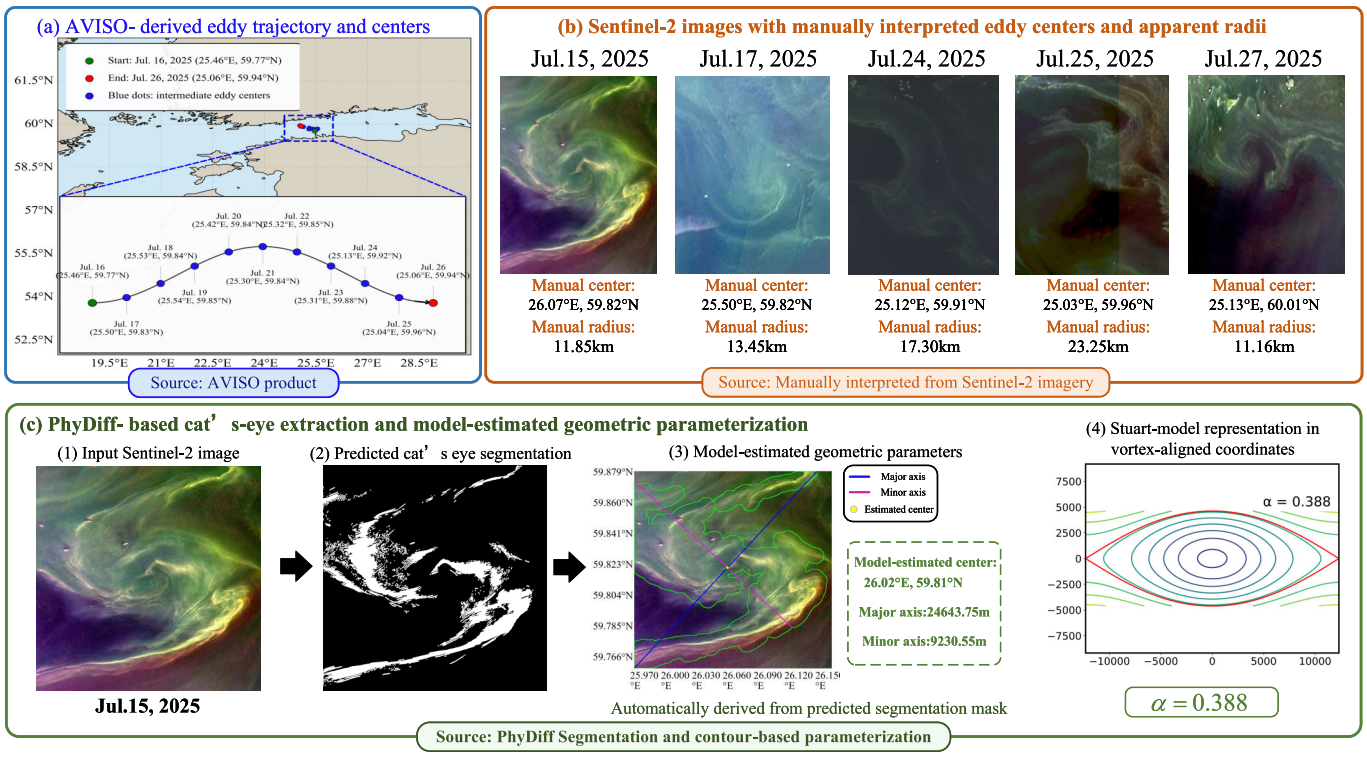


Fig. 13. Multisource eddy tracking and PhyDiff-based geometric characterization of a representative cat's-eye pattern. (a) AVISO-derived eddy trajectory and center coordinates from July 16 to 26, 2025. (b) Multidate Sentinel-2 observations with manually interpreted apparent centers and radii. (c) PhyDiff analysis of the July 15 observation, including the input image, predicted mask, contour-based geometric parameters, and the corresponding Stuart-model representation with $\alpha = 0.388$. The Stuart-model contours are displayed in a vortex-aligned coordinate system.

Algorithm 1 Contour-Based Geometric Parameterization

Require: Predicted probability map P
Ensure: Geometric descriptor set $\Gamma = \{c, v_{\text{maj}}, L_{\text{maj}}, W_{\text{true}}\}$
Threshold P to obtain the binary vortex mask Ω;
Apply morphological closing to refine Ω;
Extract the dominant vortex contour $Y = \{p_i\}_{i=1}^{N}$;
Estimate the reference center c using Eq. (23);
Compute the covariance matrix of the centered contour using Eq. (24);
Perform principal component analysis (PCA) to obtain the principal direction v_{maj};
Construct the contour-clipped longitudinal chord along v_{maj};
Compute the longitudinal extent L_{maj} using Eq. (26);
Compute the orthogonal direction v_{wid} using Eq. (27);
Construct the contour-clipped transverse chord along v_{wid};
Compute the transverse extent W_{true} using Eq. (30);
return Γ;

interpreted center in Fig. 13(c) and the model-estimated center in Fig. 13(c) are obtained using different procedures; their spatial proximity therefore provides only a qualitative consistency check rather than an independent quantitative validation.

Finally, the extracted geometric parameters are subsequently related to the idealized Stuart cat's-eye formulation. For this case, the estimated length and width yield a Stuart-model parameter of $\alpha = 0.388$. The final subpanel shows the corresponding idealized Stuart-model contour representation. These contours are displayed in a vortex-aligned coordinate system, in which the horizontal direction is aligned with the model-estimated principal axis. Their visual orientation therefore differs from the longitude–latitude coordinate system used in the preceding subpanel. This representation provides a geometry-constrained interpretation within the Stuart framework and should not be regarded as a reconstruction of the observed velocity, vorticity, or complete dynamical field.

V. Conclusion and Discussion

This study presents PhyDiff, a physics-prior diffusion model for cat's-eye vortex pattern identification in satellite imagery. By combining the synthetic CATSIM dataset with a Fourier-guided U-Net, the proposed method improves the recovery of thin and spatially continuous filaments on the synthetic benchmark. When applied to Sentinel-2 observations without additional fine-tuning, PhyDiff also produces visually coherent segmentation results. Because pixel-level ground-truth masks are unavailable for the real observations, however, these experiments demonstrate qualitative transferability rather than quantitatively verified accuracy.

The outputs of PhyDiff should be interpreted as morphological and geometric information rather than as direct dynamical measurements. Based on the predicted masks, the proposed parameterization procedure estimates the contour-based center, major-axis length, minor-axis width, and MAR of cat's-eye structure. Therefore, the inferred Stuart parameter is regarded as a morphology-based Stuart equivalent variable, used to evaluate the compatibility of the adopted idealized morphology. It

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does not constitute a direct estimate of velocity, kinetic energy, or other dynamic quantities. Similarly, contour visualization provides an internal assessment of geometric plausibility under the adopted priors, rather than a reconstruction or independent validation of the actual oceanic flow field. The representative AVISO–Sentinel-2 case further illustrates how PhyDiff can be incorporated into a multisource analysis workflow.

PhyDiff addresses a different task from conventional eddy-detection methods, which generally identify candidate eddies or complete eddy regions using bounding boxes, closed SSH contours, velocity-based criteria, or region-level masks. PhyDiff instead extracts internal filamentary structures within independently localized regions. A rigorous comparison therefore requires a unified expert-annotated dataset containing both eddy dynamics and filament-level labels, which is currently unavailable. Future work will focus on constructing such datasets, integrating PhyDiff with established detection and tracking methods, reducing the cross-modal gap between synthetic SAR-like samples and optical observations, and validating the extracted morphology using independent SSH [48], velocity, Okubo–Weiss parameter [49], and finite-size Lyapunov exponent diagnostics [50].

Several limitations remain. First, the idealized Stuart formulation and simplified synthetic image-generation procedure cannot fully represent the complexity of real ocean turbulence, biological variability, atmospheric contamination, or sensor-specific radiometric responses. Second, transferring a model trained on synthetic SAR-like samples to Sentinel-2 optical observations introduces a substantial cross-modal domain gap. Third, the absence of expert pixel-level annotations limits quantitative validation on real satellite observations. Finally, the current sequential segmentation and parameterization procedure may propagate segmentation uncertainty into the derived geometric descriptors.

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